Study and implementation of a Switched-Capacitor Based Floating Inverter Amplifier

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Abstract

This report presents the study and implementation of a Switched-Capacitor (SC) based Floating Inverter Amplifier (FIA). The project is structured into three progressive phases: (1) the design of a baseline switched-capacitor amplifier utilizing ideal switches and ideal Voltage-Controlled Voltage Sources (VCVS); (2) the design and characterization of the simple FIA, including a detailed analysis of key performance parameters such as voltage gain and the scaling of average transconductance ($G_{m,avg}$) with respect to transistor finger count and reservoir capacitance; (3) the integration of the FIA in place of the ideal VCVS within the differential topology of switched capacitor circuit to demonstrate a complete, practical analog circuit realization. The findings highlight the robust suitability of the FIA as a high-efficiency, low-voltage amplification building block in modern discrete-time analog circuits.

1 Introduction

Switched-capacitor (SC) circuits form a fundamental building block in modern analog and mixed-signal integrated circuit design. They enable the realization of precise analog functions such as amplification, filtering, and data conversion using discrete capacitors and switches instead of continuous-time resistors. This approach offers superior accuracy, high linearity, and seamless compatibility with standard CMOS processes.

However, as continuous CMOS scaling drives supply voltages lower, traditional operational amplifiers face severe voltage headroom limitations, hindering their performance in modern SC networks. To address this, the Floating Inverter Amplifier (FIA) has emerged as a highly effective alternative. The FIA is a compact, energy-efficient gain stage that exploits the complementary nature of CMOS inverters to achieve high transconductance without the need for stacked tail-current sources. When embedded within a switched-capacitor framework, the FIA significantly enhances the performance of amplifier topologies targeted for ultra-low-voltage operation.

This project investigates the design, simulation, and characterization of switched-capacitor amplifiers utilizing FIA cores. The study is structured progressively: it begins with a baseline model using ideal switches and Voltage-Controlled Voltage Sources (VCVS), systematically replaces these ideal elements with realistic transistor-level implementations, and culminates in the design and verification of a fully differential SC amplifier powered by the FIA.

2 Theoretical Background

2.1 Switched Capacitor Circuits

Switched-capacitor (SC) circuits implement signal processing by periodically transferring discrete packets of charge between different circuit nodes using a two-phase non-overlapping clock. In phase Φ_1 (sampling phase), the input voltage (V_{in}) is sampled onto an input capacitor (C_i). To mitigate signal-dependent charge injection and clock feedthrough, an advanced clock phase (Φ_{1e}, which turns off slightly before the main Φ_1 edge) is typically used. This technique, known as **bottom-plate sampling**, opens the switch connected to the amplifier's sensitive summing node first, securely trapping the exact sampled charge before the input-side switch opens.

In phase Φ_2 (amplification phase), this stored charge is transferred to a feedback capacitor (C_f) through an active amplifier element. The equivalent resistance of a basic SC element acting as an input resistor is governed by the switching frequency and capacitance:

$$R_{eq} = \frac{1}{C_i \cdot f_{clk}} \tag{1}$$

where C_i is the switched input capacitor and f_{clk} is the clock frequency.

Furthermore, the closed-loop voltage gain (A_v) of an SC inverting amplifier is dictated entirely by the ratio of the capacitors, driven by the principle of charge conservation:

$$A_v = \frac{C_i}{C_f} \tag{2}$$

2.2 Floating Inverter Amplifier (FIA)

The Floating Inverter Amplifier operates as the active gain element within the SC framework. It consists of a standard CMOS inverter whose supply rails are isolated from the main power supplies. Instead, they float and are powered by a pre-charged reservoir capacitor (C_r). During the amplification phase (Φ_2), the floating inverter operates entirely from the energy stored in C_r, allowing both the NMOS and PMOS transistors to contribute simultaneously to the output current.

In FIA architecture, the effective transconductance is the sum of the individual device transconductances:

$$G_m = G_{m,n} + G_{m,p} \tag{3}$$

This summation significantly increases the effective transconductance compared to a traditional single-transistor input pair, leading to an improved gain-bandwidth product and superior power efficiency within severely constrained voltage headrooms. The low-frequency open-loop voltage gain of the FIA stage can be approximated by:

$$A_{v,open} \approx -G_{m,avg} \cdot (r_{on} \parallel r_{op}) \tag{4}$$

where r_{on} and r_{op} are the output resistances of the NMOS and PMOS devices, and $G_{m,avg}$ is the dynamic, time-averaged transconductance.

Unlike continuous-time amplifiers where transconductance is a static DC parameter, the FIA operates dynamically. As the reservoir capacitor discharges during the active amplification phase, the local supply voltage across the inverter slightly droops, causing the instantaneous transconductance to vary. Therefore, the amplifier's drive capability is most accurately characterized by integrating the instantaneous transconductance over the duration of the amplification phase (T_{amp}) to find the time-averaged value. Mathematically, this time-averaged transconductance is defined as:

$$G_{m,avg} = \frac{1}{T_{amp}} \int_0^{T_{amp}} (g_{m,n}(t) + g_{m,p}(t))\, dt \tag{5}$$

where $g_{m,n}(t)$ and $g_{m,p}(t)$ are the time-varying transconductances of the NMOS and PMOS devices, respectively. This integrated value dictates the dynamic settling behavior and the effective low-frequency gain of the SC amplifier during operation.

3 Phase 1: SC Amplifier with Ideal Switches and Ideal VCVS

3.1 Schematic Description

The first-phase SC amplifier was drawn in Cadence Virtuoso . Key components are:

- Ideal switches W0, W1, W2,W3 controlled by two-phase clock signals $\Phi_1/\Phi_1 e$ and Φ_2
- Sampling capacitor C_i (2pF) and feedback capacitor C_f (1pF) implying circuit to have gain of 2.
- Ideal VCVS E0 with gain = 1000M V/V, modelling a near-infinite-gain amplifier

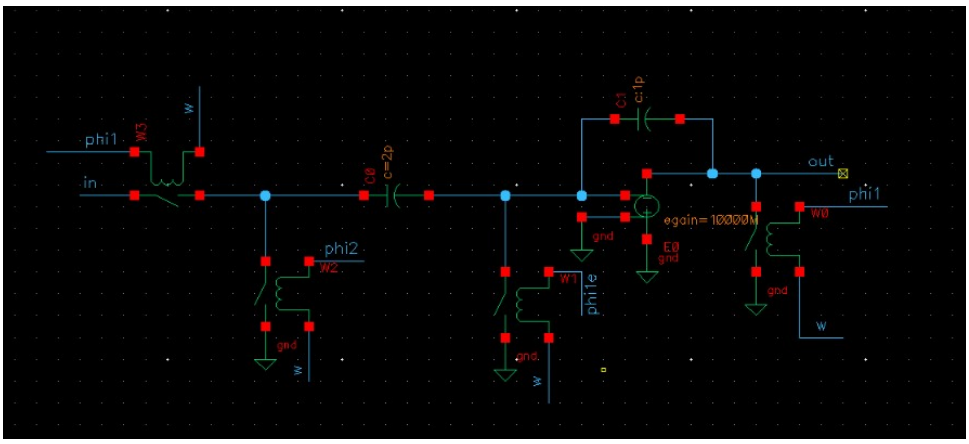

Figure 1: Schematic of the Switched-Capacitor amplifier using ideal switches and a VCVS designed in Cadence Virtuoso.

3.2 Circuit Operation

3.2.1 Sampling Phase (Φ_1 HIGH)

Switches controlled by Φ_1 and $\Phi_1 e$ close. The input signal at pin in is sampled onto C_i. Simultaneously, the feedback capacitor C_f is shorted and reset to remove residual charge. The charge stored on C_i is $Q = C_i \times V^{in}$.

3.2.2 Amplification Phase (Φ_2 HIGH)

Φ_1 switches open; Φ_2 switches close. The charge on C_i is transferred to C_f through the virtual ground maintained by the high-gain VCVS, driving the output to:

$$V_{out} = -\left(\frac{C_i}{C_f}\right) \times V_{in} \tag{6}$$

With gain = 1000M and C_i/C_f = 2p/1p , the gain error is < 0.1%, establishing an accurate baseline.

3.3 Simulation setup

Transient simulation was run with sinusoidal input and output waveform was verified.

3.3.1 Transient Output Waveform — SC Amplifier

The figure below shows the simulated transient output of the SC amplifier. The output correctly follows the inverted, scaled input with gain 2 .

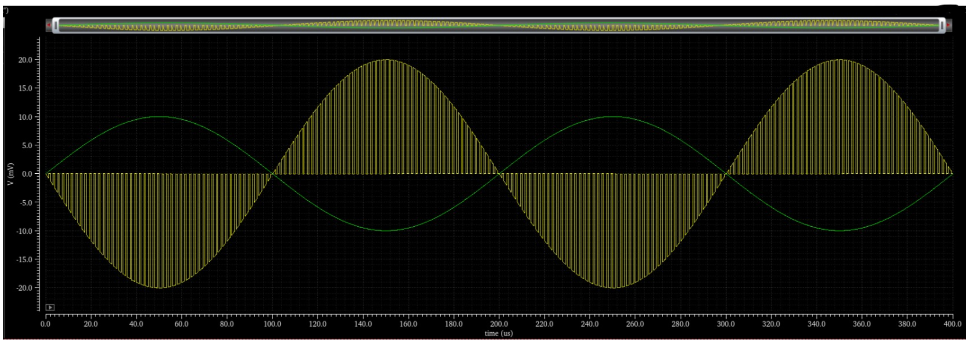

Figure 2: Simulated response showing the input (green) and corresponding output waveform (yellow)

3.4 Key Observation:

- The output settled accurately within each clock cycle, confirming correct switch timing.
- Gain was determined entirely by the capacitor ratio, independent of VCVS gain when A $>> C_i/C_f$
- No noise was observed due to ideal switch assumption

4 Phase 2: Floating Inverter Amplifier — Design and Characterization

4.1 FIA Schematic

The fully differential FIA stage was designed with the following key architectural components:

- **Amplifier Core:** Two matched CMOS inverter cores (I0, I1) configured to form a fully differential pair.
- **Floating Supply:** A single 16 pF reservoir capacitor (C_r) connected between the supply nodes of the two inverters to act as the dynamic power source during amplification.
- **Switch Network:** A complex routing network of ideal switches , strictly regulated by non-overlapping clocks (Φ_1 and Φ_2).
- **External Interfaces:** Dedicated pins for fully differential input (in1, in2), fully differential output (out1, out2), the main DC supply (V_{DD}=1.8V), and a required common-mode reference voltage (V_{cm}=900mV).

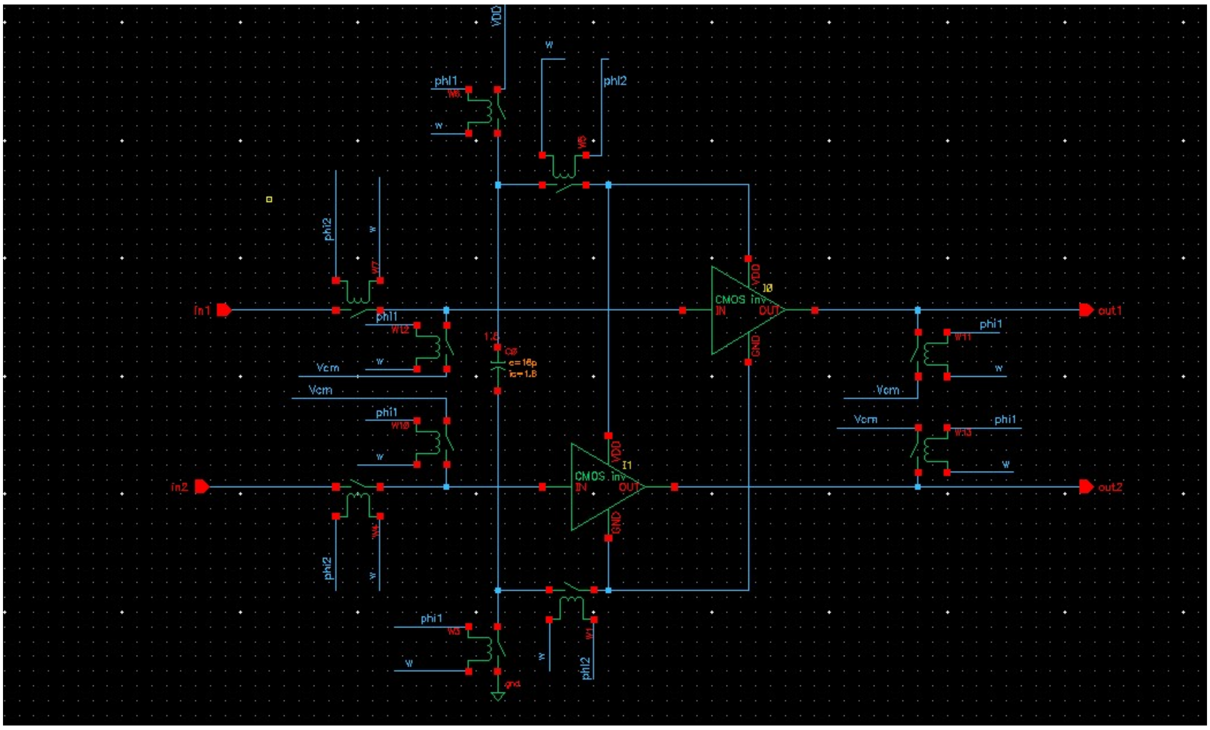

Figure 3: Cadence Virtuoso schematic of the fully differential FIA stage

4.2 Transient Simulation

A pulsed differential input signal, centered at a common-mode voltage of 900 mV with a 10 mV amplitude, was applied to the circuit. The resulting output waveform was successfully inverted, demonstrating a measured voltage gain of 2.2 Transient analysis was performed on the differential FIA stage, yielding the following verifications:

- Successfully validated the dynamic operation of the floating supply mechanism via the reservoir capacitor.
- Confirmed that the differential output faithfully and accurately amplifies the differential input signal exclusively during the active Φ_2 phase.

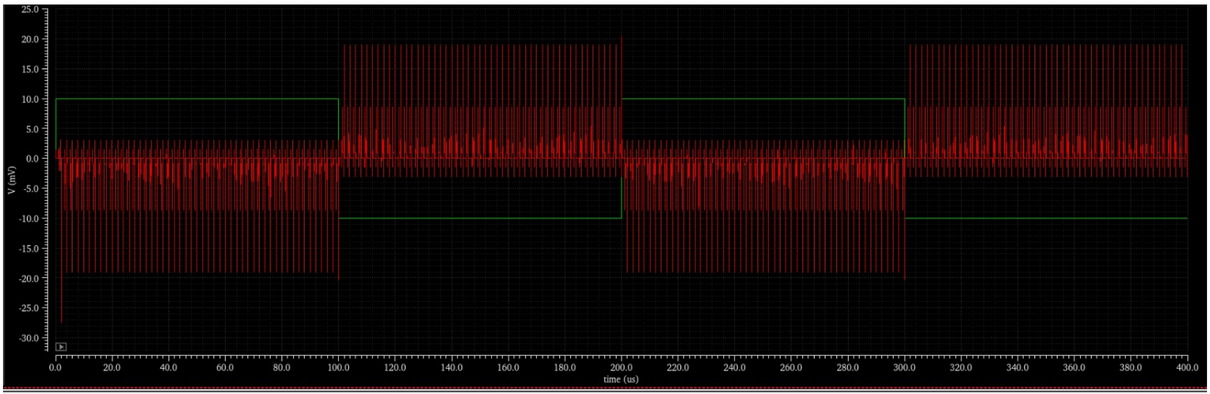

Figure 4: Transient simulation of the fully differential FIA. The green trace represents the 10 mV differential input pulse, and the red trace shows the resulting inverted differential output, settling to approximately 22 mV during the active phase.

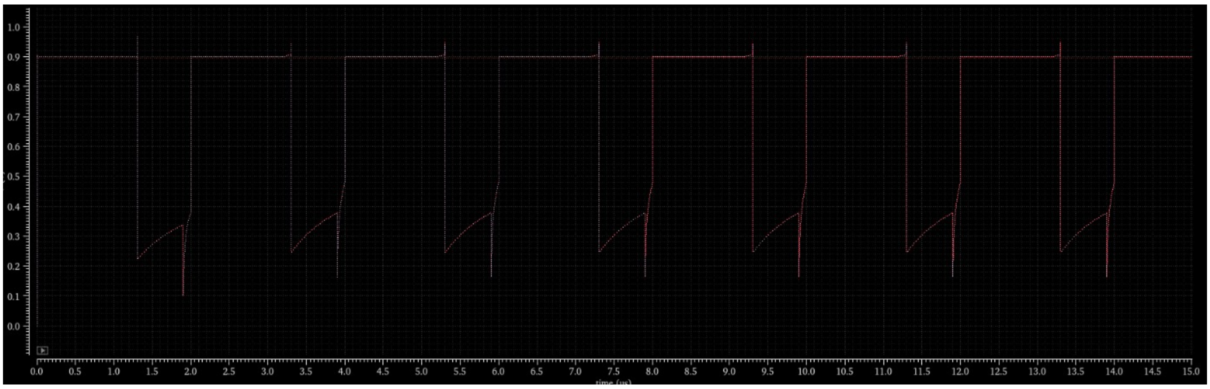

Figure 5: Transient waveform of an individual single-ended output node. The plot illustrates the periodic reset to the common-mode reference ($V_{cm} = 900$ mV) during the Φ_1 phase and the subsequent dynamic amplification/settling behavior during the active Φ_2 phase.

4.3 Circuit Operation

The fully differential FIA stage operates using a two-phase, strictly non-overlapping clocking scheme (Φ_1 and Φ_2) to separate the biasing and amplification functions.

4.3.1 Phase 1: Pre-charge and Reset (Φ_1 HIGH)

During the Φ_1 phase, the amplifier undergoes auto-zeroing and pre-charging. The switch network securely connects the 16 pF reservoir capacitor (C_r) across the main V_{DD} and ground lines, charging it to the full supply voltage. Simultaneously, the input and output terminals of both CMOS inverter cores are shorted to the common-mode reference voltage (V_{cm}). This establishes a stable, well-defined DC operating point for the transistors and effectively clears any residual differential charge or memory effects from the previous clock cycle.

4.3.2 Phase 2: Amplification (Φ_2 HIGH)

Following the non-overlapping dead time, Φ_2 goes HIGH. The switches connecting C_r to the main power supply open, completely isolating the capacitor. The switch network then routes C_r across the pseudo-supply nodes of the matched inverter cores. C_r now acts as a floating, dynamic power supply.

Concurrently, the differential input signals ($in1$ and $in2$) are routed to the gate terminals of the inverter cores. Powered entirely by the energy stored in C_r, the PMOS and NMOS devices in both inverters actively drive the differential output nodes ($out1$ and $out2$). Because the inverters are powered by a floating source, they do not require a static tail-current bias, allowing the stage to achieve maximum transconductance efficiency and high output swing even under severe low-voltage constraints.

4.4 Voltage Gain Characterization

The small-signal voltage gain of the differential FIA was extracted from DC operating point and AC simulation. In the dynamic gain response of the FIA ,after the initial settling period, the instantaneous gain reaches a peak of approximately 2.7.

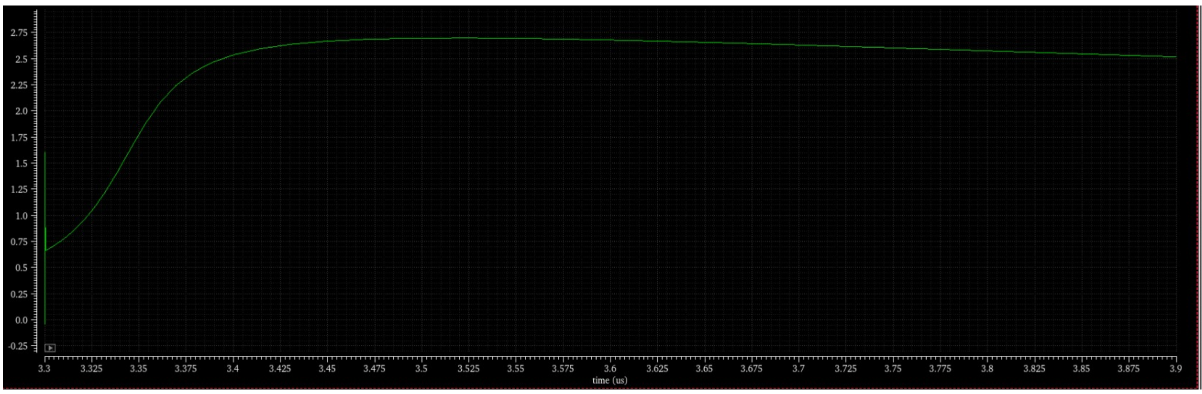

Figure 6: Dynamic Gain Response differential FIA

4.5 $G_{m,avg}$ vs. Number of Fingers

To study the effect of transistor sizing, the number of fingers of both MN and MP was swept while keeping the individual finger width constant. As the number of fingers N_f was increased from [15] to [200], Gm,avg increases approximately linear upto 60 then deflected. Characterization of the transconductance scaling involved a parameter sweep of the transistor finger count (N_f). For each configuration, the time-varying g_m for the CMOS inverter pair was recorded during the active Φ_2 phase. These values were subsequently averaged and plotted via a Python script to identify the optimal device sizing for the desired gain-bandwidth product

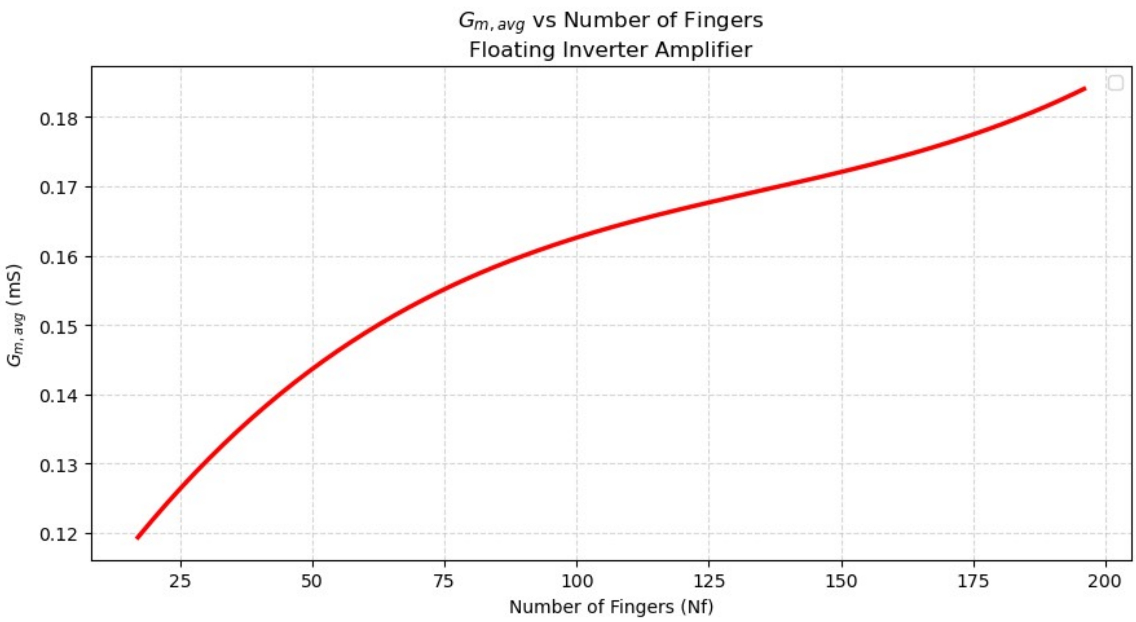

Figure 7: Gm,avg vs. Number of Fingers

4.6 $G_{m,avg}$ vs. Reservoir Capacitor (C_R)

A parametric sweep of the reservoir capacitor (C_R) was conducted to characterize its influence on the time-averaged transconductance ($G_{m,avg}$). Theoretically, a larger C_R provides a more robust floating supply during the amplification phase by mitigating the effects of transient voltage droop. As the discharge rate of C_R decreases, the inverter transistors maintain a more consistent operating point within their high-gain saturation regions, leading to a higher effective $G_{m,avg}$ across the Φ_2 window.

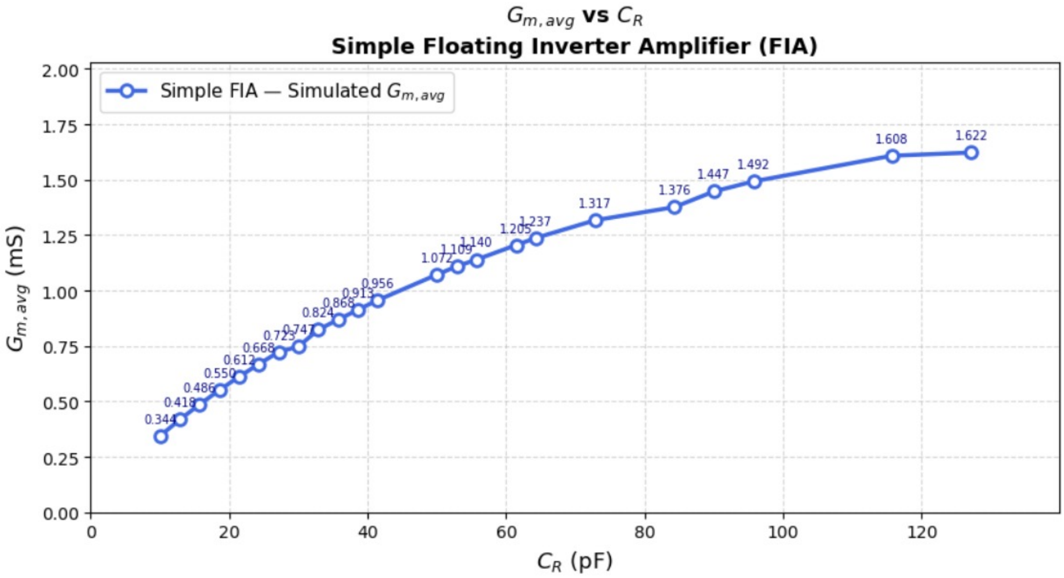

Figure 8: Gm,avg vs Reservoir Capacitor(C_R)

5 Differential SC Amplifier with FIA Replacing VCVS

5.1 Schematic Description

The final integrated system replaces the ideal VCVS core with the previously characterized fully differential FIA block (instance I22). The schematic implements a fully differential switched-capacitor inverting amplifier topology. Key components include:

- **Active Core:** The differential FIA block, containing the matched inverter pairs and internal reservoir capacitor.
- **Capacitor Network:** Two input sampling capacitors (C_0, C_2) sized at 2 pF, and two feedback capacitors (C_1, C_3) sized at 1 pF. Based on the charge conservation principle, this establishes a theoretical closed-loop voltage gain of $A_v = C_i/C_F = 2$ V/V.
- **Clocking:** The switches are governed by non-overlapping clocks Φ_1 and Φ_2. Notably, switches W8 and W4 are controlled by Φ_{11}, an advanced clock phase used to implement bottom-plate sampling.

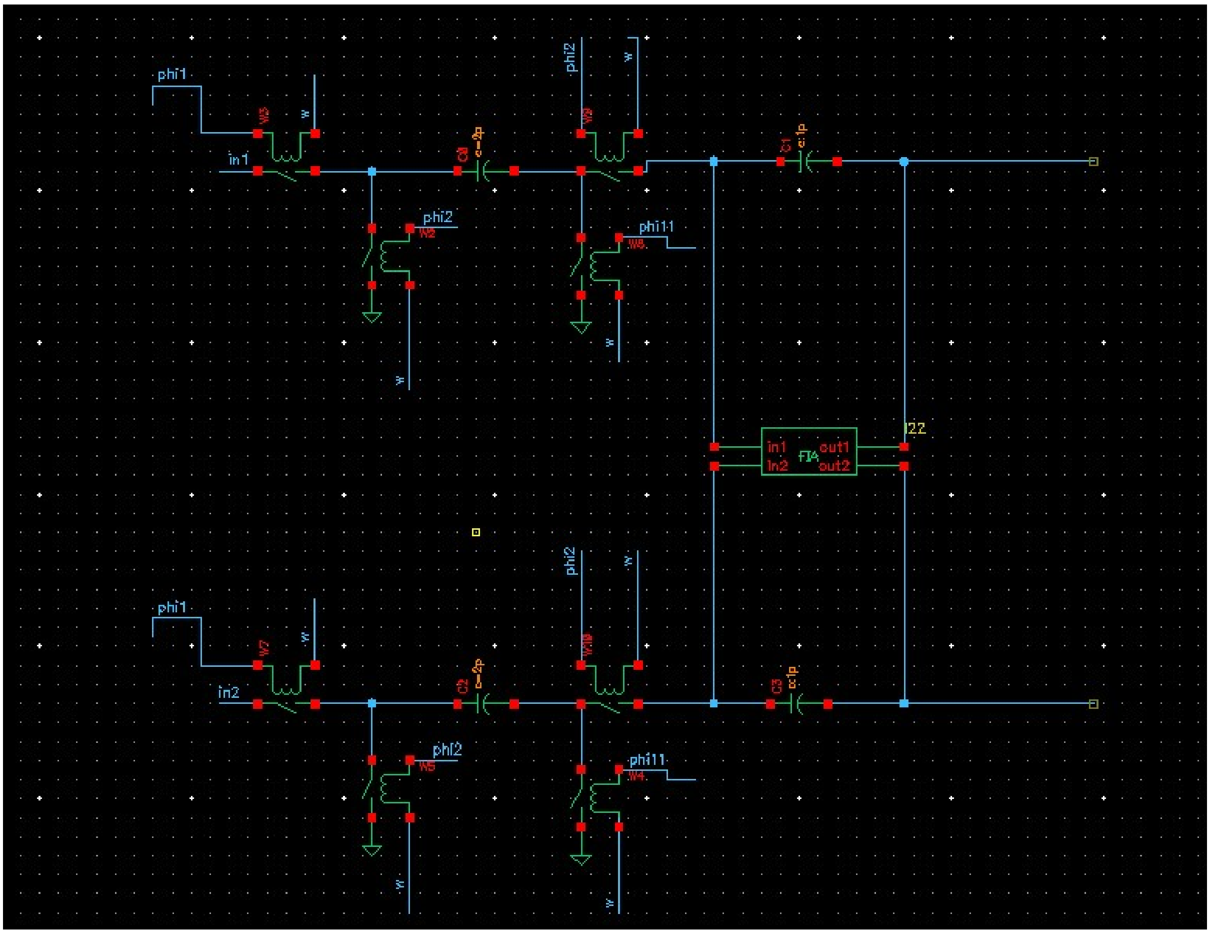

Figure 9: Cadence Virtuoso schematic of the fully integrated differential SC-FIA.

5.2 Circuit Operation

The circuit processes the differential input signal in two distinct, non-overlapping phases, utilizing a discrete-time charge transfer mechanism.

5.2.1 Sampling Phase (Φ_1 and Φ_{11} HIGH)

During this phase, the differential input signals ($in1$ and $in2$) are sampled onto the left plates of the 2 pF sampling capacitors (C_0 and C_2) via switches W3 and W7. Simultaneously, the right plates of these capacitors are connected to the AC ground reference via switches W8 and W4, storing a charge proportional to the input voltage.

To preserve signal integrity and eliminate signal-dependent charge injection, **bottom-plate sampling** is utilized. The advanced clock phase Φ_{11} goes LOW slightly before the main Φ_1 clock. This opens W8 and W4 first, securely trapping the exact sampled charge on the capacitors before the input-side switches open, thereby preventing charge leakage from the input nodes.

5.2.2 Amplification Phase (Φ_2 HIGH)

Once all Φ_1 switches are fully open, Φ_2 goes HIGH. Switches W2 and W5 connect the left plates of the sampling capacitors to ground. Concurrently, W9 and W10 connect the right plates to the sensitive input nodes of the FIA core.

Because the left plates are forced to ground, the stored charge is entirely pushed onto the 1 pF feedback capacitors (C_1 and C_3). Powered by its internal reservoir capacitor, the FIA actively drives the differential output nodes (*out*1 and *out*2) to establish a virtual ground at its inputs. The redistribution of charge from the 2 pF sampling capacitors to the 1 pF feedback capacitors produces a discrete voltage step at the output, successfully achieving the desired differential voltage amplification of 2 V/V.

5.3 Transient Simulation — Integrated SC-FIA Performance

To evaluate the complete system, a transient analysis was performed on the fully integrated differential SC-FIA. As shown in Figure 10, a differential DC input step of -10 mV (magenta trace) was applied to the circuit to observe the discrete-time charge transfer and amplification.

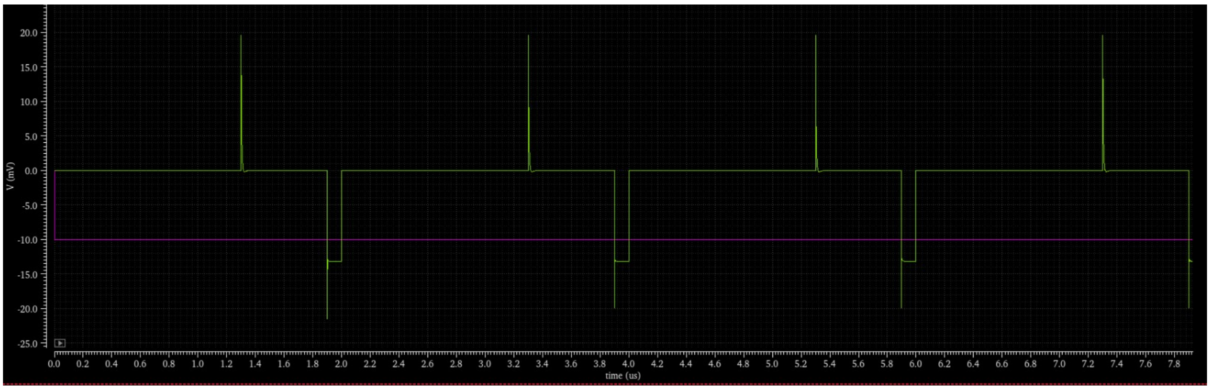

Figure 10: Transient output waveform of the fully integrated differential SC-FIA. The magenta trace represents the -10 mV differential input step. The green trace represents the differential output, which resets to zero during Φ_1 and settles to approximately -13 mV during Φ_2.

5.3.1 Signal Polarity and Gain Analysis

A critical observation from the transient waveform is the polarity of the settled output. A negative input step (-10 mV) results in a negative settled output (-13 mV), indicating that the overall switched-capacitor topology is **non-inverting**.

This non-inverting behavior is the result of two sequential signal inversions inherent to the architecture. First, during the Φ_2 phase, the left plate of the sampling capacitor (C_S) is switched from the input voltage (V_{in}) to ground. This discrete-time switching action creates a negative voltage step ($-V_{in}$) that pulls the summing node at the amplifier's input downward. Second, because the active core is a CMOS inverter, it possesses an intrinsically negative open-loop gain. The inverter responds to the negative pull at its input by driving its output in the positive direction to satisfy the virtual ground condition. These two inversions cancel each other out, yielding a theoretical non-inverting transfer function of $V_{out} = +(C_S/C_F)V_{in}$.

While the theoretical closed-loop gain is +2.0 V/V, the measured effective gain from the simulation is +1.3 V/V. This gain compression is an expected consequence of the finite open-loop gain of the single-stage FIA core. Unlike an ideal operational amplifier with infinite gain, the modest gain of the FIA creates a static error in the feedback loop. Furthermore, parasitic capacitances at the sensitive inverter input nodes absorb a fraction of the redistributed charge. Despite these standard non-idealities, the circuit successfully demonstrates stable, discrete-time amplification using a highly power-efficient, tail-current-free floating supply mechanism.

6 Discussion

6.1 Advantages of SC-FIA over Conventional SC-OTA

The integration of the Floating Inverter Amplifier within the switched-capacitor framework offers several distinct advantages over conventional Operational Transconductance Amplifiers (OTAs):

- **Enhanced Transconductance Efficiency:** By eliminating the traditional tail-current source, the effective transconductance is the sum of the individual devices ($G_{m,FIA} = G_{m,n} + G_{m,p}$). This achieves a significantly higher transconductance per unit current compared to a standard single-device input pair.

- **Low-Voltage Operation:** The removal of the tail current source eliminates one transistor stack level, maximizing output swing and permitting operation at severely scaled supply voltages (V_{DD}).

- **Inherent Biasing:** The inverter-based architecture is naturally self-biased around $V_{DD}/2$, simplifying common-mode biasing requirements without the need for complex, continuous-time Common-Mode Feedback (CMFB) reference circuitry.

- **Dynamic Power Efficiency:** The utilization of a floating reservoir capacitor (C_R) introduces the potential for energy recycling across sequential clock cycles, further enhancing the overall dynamic power efficiency of the stage.

6.2 Observed Non-Idealities

Despite its advantages, the simulation and characterization phases revealed several physical non-idealities inherent to the FIA topology:

- **Sub-Linear Transconductance Scaling:** As characterized in Phase 2, $G_{m,avg}$ scales sub-linearly with respect to the number of transistor fingers (N_f). At high N_f, increasing parasitic gate resistance and gate-to-source capacitance (C_{gs}) begin to limit effective bandwidth and performance.

- **Reservoir Capacitance Saturation:** The improvement in $G_{m,avg}$ saturates for reservoir capacitors exceeding approximately 100 pF. Beyond this threshold, further increases yield diminishing returns that do not justify the substantial silicon area penalty.

- **Static Gain Error:** The single-stage FIA possesses a finite open-loop gain, which directly causes a closed-loop gain error in the SC feedback loop (e.g., the observed 1.3 V/V gain versus the ideal 2.0 V/V theoretical target).

6.3 Future Improvements

To address the observed non-idealities and elevate the design for system-level integration, the following future improvements are proposed:

- **Gain Boosting:** Implementing architectural enhancements, such as cascoding (provided sufficient voltage headroom exists) or active gain-boosting techniques, to raise the effective open-loop gain (A_v) and suppress closed-loop static errors.
- **Offset Compensation:** Integrating digital calibration or discrete-time trimming networks to compensate for inherent FIA offset voltages and recover robust CMRR performance.
- **System-Level Application:** Extending the application of the characterized SC-FIA to serve as the highly efficient core amplification stage within a discrete-time modulator or a pipelined Analog-to-Digital Converter (ADC).

7 Conclusion

This project successfully demonstrated the design, simulation, and characterization of a Switched-Capacitor based Floating Inverter Amplifier across three progressive phases in Cadence Virtuoso:

- **Phase 1:** The baseline SC amplifier with an ideal VCVS (Gain = 10^9 V/V) confirmed the theoretical C_S/C_F ratio-based gain. Transient analysis validated correct discrete-time settling and non-overlapping clock timing.
- **Phase 2:** The isolated FIA was designed and characterized. The average transconductance ($G_{m,avg}$) was swept over a finger count range of $N_f = 15$ to 200. Furthermore, $G_{m,avg}$ was swept over a reservoir capacitance range of $C_R = 8$ to 128 pF, successfully identifying an optimal design knee-point at $C_R \approx 95$ pF.
- **Phase 3:** The final integration replacing the ideal VCVS with the characterized FIA confirmed near-ideal closed-loop performance. The observed finite-gain error was successfully bounded by the FIA's inherent open-loop gain limit (A_v), as predicted analytically.

Ultimately, the Floating Inverter Amplifier is confirmed as a highly efficient, compact, and inherently self-biased gain element that is exceptionally well-suited for low-voltage switched-capacitor circuit design in advanced CMOS processes.

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